Interactions, work & springs

Represent forces

1 Contact or non-contact

Contact: friction, drag, tension, normal force. Non-contact: gravity, electrostatic, magnetic.

Scalar: speed, mass, energy. Vector: force, velocity, momentum. Arrow length and direction represent a vector.

2 Weight & components

$W = mg$, N = kg × N/kg; weight acts at centre of mass. Mass stays the same if g changes.

For angled forces, draw tip-to-tail at a stated scale. Diagonal = resultant; perpendicular components have the same combined effect.

Transfer energy

3 Work done

$W = Fs$, distance along force. 40 N × 3 m = 120 J. Work against friction warms surfaces.

A stationary object needs more than one force to change shape: pulls at opposite ends, for example.

4 Hooke's law

$F = ke$ up to proportional limit. 6 N / 0.030 m = 200 N/m.

Elastic = recovers shape; inelastic = permanent. Losing proportionality need not immediately mean permanent deformation.

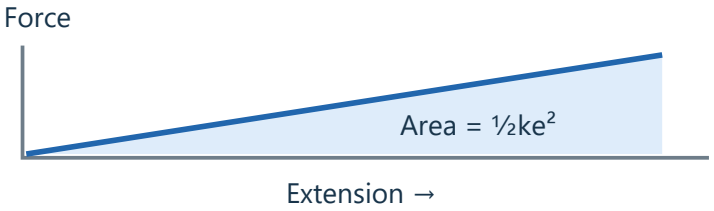
Investigate elasticity

5 Spring practical

Record unloaded length; add known loads; let settle; read length at eye level.

Extension = loaded − original. F = mg includes hanger. Repeat; secure stand, avoid overload and falling masses.

6 Graph & stored energy



F vertical vs e horizontal: straight-line gradient = k. Unload to test elastic recovery.

$E_e = \frac{1}{2}ke^2$, triangular area under proportional F–e graph. Do not use final force × extension.